Name: _____ Date: _____

PS-04: Integration Techniques - Substitutions

Due: Wednesday, 2026 November 4
• Integration by Substitution

Directions: This will be graded for correct work and answers.

Perform the indicated integrations. Provide exact numerical answers for relevant problems, not decimal approximations.

1. $\int_0^1 \frac{6e^x}{2+3e^x}\,dx$
2. $\int_0^1 x\sqrt{1-x^2}\,dx$
3. $\int \frac{1}{x^2+4}\,dx$
4. $\int \frac{2x+1}{x^2+9}\,dx$
5. $\int \frac{2t^2}{2t^2+1}\,dt$
6. $\int \frac{4}{\sqrt{2t+1}}\,dt$
7. $\int \frac{x^3+7x}{x-1}\,dx$
8. $\int \frac{\sin\left[\ln(4x^2)\right]}{x}\,dx$
9. $\int \frac{\sec^2(\ln(x))}{2x}\,dx$
10. $\int \frac{5e^x}{\sqrt{1-e^2x}}\,dx$
11. $\int \frac{x}{x^2+4}\,dx$

Solution 1

Solution 1.1

Desmos Let $\varepsilon > 0$ and choose $M = 1 + \dfrac{1}{\varepsilon} > 1$. Then, if $t > M$, we have

$$\begin{aligned}
|f(t) - L| &= \left| \frac{t^2}{t - t^2} - (-1) \right| \\
&= \left| \frac{t^2}{t - t^2} + 1 \right| \\
&= \left| \frac{t^2 + (t - t^2)}{t - t^2} \right| \\
&= \left| \frac{t}{t(1-t)} \right| && \text{since } t > M \geq 0 \\
&= \frac{1}{|1 - t|} \\
&= \frac{1}{|t - 1|} \\
&= \frac{1}{t - 1} && \text{since } t > M > 1 \\
&< \frac{1}{M - 1} && \text{since } \frac{1}{x-1} \text{ is decreasing for } x > 1 \\
&= \frac{1}{(1 + \frac{1}{\varepsilon}) - 1} \\
&= \varepsilon \qquad \square
\end{aligned}$$

Solution 1.2

Let $\varepsilon > 0$ and choose $M = \ldots$. Then, if $t > M$, we have

$$\begin{aligned}
|f(t) - L| &= \left| \frac{t^2}{t - t^2} - (-1) \right| \\
&= \left| \frac{t^2}{t - t^2} + 1 \right| \\
&= \left| \frac{t^2 + (t - t^2)}{t - t^2} \right| \\
&= \left| \frac{t}{t(1-t)} \right| && \text{since } t > M \geq 0 \\
&= \frac{1}{|1 - t|} \\
&= \frac{1}{|t - 1|} \\
&= \frac{1}{t - 1} && \text{since } t > M > 1 \\
&< \frac{1}{M - 1} && \text{since } \frac{1}{x-1} \text{ is decreasing for } x > 1 \\
&= \frac{1}{(1 + \frac{1}{\varepsilon}) - 1} \\
&= \varepsilon \qquad \square
\end{aligned}$$

Solution 2

Solution 2.1

Let $\varepsilon > 0$ and choose $M = \max\left(16, \frac{48}{\varepsilon}\right)$. Then, if $x > M$, we have

$$\begin{aligned} |f(x) - L| &= \left| \frac{3x^2}{x^2 - 8x + 15} - 3 \right| \\ &= \left| \frac{3x^2 - 3(x^2 - 8x + 15)}{x^2 - 8x + 15} \right| \\ &= \left| \frac{24x - 45}{x^2 - 8x + 15} \right| \\ &= \frac{|24x - 45|}{|x^2 - 8x + 15|} \end{aligned}$$

Considering the numerator and denominator separately, we have

$$|24x - 45| > 0 \quad \text{if } x > 2$$

and

$$\left|x^2 - 8x + 15\right| = |(x-3)(x-5)| > 0 \quad \text{if } x > 5$$

Let's assume, then, that $M \geq 5$. Therefore,

$$\begin{aligned} |f(x) - L| = \ldots &= \frac{|24x - 45|}{|x^2 - 8x + 15|} \\ &= \frac{24x - 45}{x^2 - 8x + 15} \\ &< \frac{24x}{x^2 - 8x + 15} \end{aligned}$$

Let's bound the denominator:

$$x^2 - 8x + 15 > \frac{1}{2}x^2 \iff \frac{1}{2}x^2 - 8x + 15 > 0$$

To find the zeros of the above quadratic expression, use the quadratic formula:

$$x = \frac{16 \pm \sqrt{156 - 4(30)}}{2} = \frac{16 \pm \sqrt{244}}{2} = \frac{16 \pm 2\sqrt{61}}{2} = 8 \pm \sqrt{61}$$

Now, the larger root is $8 + \sqrt{61} < 16$. Therefore, for $x > 16$, we have

$$x^2 - 8x + 15 > \frac{1}{2}x^2$$

Therefore, we'll assume $M \geq 16$ and can say

$$\begin{aligned} |f(x) - L| = \ldots &= \frac{24x - 45}{x^2 - 8x + 15} \\ &< \frac{24x}{\frac{1}{2}x^2} \\ &= \frac{48}{x} \\ &= \frac{48}{48/\varepsilon} \\ &= \varepsilon \qquad \square \end{aligned}$$

Solution 2.2

Setup:

$$\left|\frac{3x^2}{x^2-8x+15}-3\right| = \ldots = 3\left|\frac{8x-15}{x^2-8x+15}\right|$$

See the solution here]
